

# A Comparative Study of Apriori and CARTS (Contextual Awareness Recommendation and Transaction System) for Grocery Recommendation Systems

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**Abstract**—Recommendation systems are critical tools for retail businesses, enhancing both sales and customer satisfaction. Traditional market basket analysis using the Apriori algorithm primarily considers frequency-based metrics such as support and confidence, neglecting financial performance. This paper introduces CARTS (Contextual Awareness Recommendation and Transaction System), which integrates profitability metrics alongside frequency measures. By applying Z-score normalization, CARTS balances item frequency with profit and margin, enabling profit-aware recommendations. We present detailed workflows for both Apriori and CARTS, provide example outputs for 1-itemset and 2-itemset rules, compare their effectiveness, and discuss advantages, limitations, and practical implications for retail businesses.

**Index Terms**—Recommendation Systems, Apriori, CARTS, Market Basket Analysis, Z-score Normalization, Profitability

## I. INTRODUCTION

Understanding customer purchasing patterns is essential for modern retailers to drive revenue growth and enhance customer satisfaction. Association rule mining, especially the Apriori algorithm, is widely used in market basket analysis to discover frequently co-purchased items. However, Apriori emphasizes only transaction frequency, ignoring profitability, which can lead to suboptimal recommendations from a business perspective. For instance, a low-margin staple item may dominate recommendations due to its high frequency, while high-margin items are overlooked.

To address this gap, we propose the CARTS framework, which extends traditional Apriori by incorporating financial attributes such as profit percentage and margin into rule evaluation. CARTS uses Z-score normalization to combine frequency-based metrics with profitability, generating recommendations that are both customer-centric and profit-oriented. This paper compares the workflows and outcomes of Apriori and CARTS, highlighting CARTS' ability to elevate profitable yet less frequent item associations.

## II. BACKGROUND

### A. Market Basket Analysis

Market Basket Analysis (MBA) is a data mining technique aimed at uncovering associations between items purchased

together in transactions. MBA helps retailers understand buying behavior, optimize store layouts, and design targeted promotions. Example insight: “customers who buy tea often purchase biscuits as well.”

### B. Apriori Algorithm

The Apriori algorithm identifies frequent itemsets using a minimum support threshold, then generates association rules satisfying a minimum confidence. Key metrics include:

$$\text{Support}(A \rightarrow B) = \frac{|A \cup B|}{N}, \quad (1)$$

$$\text{Confidence}(A \rightarrow B) = \frac{|A \cup B|}{|A|}, \quad (2)$$

$$\text{Lift}(A \rightarrow B) = \frac{\text{Support}(A \cup B)}{\text{Support}(A) \cdot \text{Support}(B)}. \quad (3)$$

### C. Limitations of Apriori

Apriori treats all items equally, focusing solely on frequency. High-frequency low-margin items may dominate the rule ranking, while profitable combinations with lower occurrence are underrepresented. This can limit practical profitability-focused decision-making for retailers.

## III. CARTS FRAMEWORK

CARTS extends Apriori by integrating financial metrics into rule scoring, making recommendations both frequent and profitable.

### A. Why Use Z-Score in CARTS

Different attributes such as support, profit percentage, and margin exist on varying scales. Without normalization, one metric (e.g., profit%) could dominate rule evaluation unfairly. Z-score normalization standardizes features, making them comparable. By centering around the mean and scaling by standard deviation, CARTS ensures that frequency and profitability have balanced influence on final recommendations. This prevents bias and allows the framework to highlight rules that are not only frequent but also financially meaningful.

### B. Z-Score Normalization

Item attributes such as support, profit percentage, and margin exist on different scales. To standardize these features, Z-score normalization is applied:

$$Z(x) = \frac{x - \mu}{\sigma} \quad (4)$$

where  $\mu$  is the mean and  $\sigma$  is the standard deviation of the feature across all items.

### C. CARTS Scoring Function

For a rule  $A \rightarrow B$ , we define:

$$R_s(A \rightarrow B) = \text{Support}(A \cup B) \times \text{Confidence}(A \rightarrow B), \quad (5)$$

$$\text{CARTS}(A \rightarrow B) = \alpha R_s + \beta Z(\text{Profit}\%_B) + \gamma Z(\text{Margin}\%_B), \quad (6)$$

where  $\alpha + \beta + \gamma = 1$ ; typical values:  $\alpha = 0.4$ ,  $\beta = 0.2$ ,  $\gamma = 0.4$ . CARTS balances rule frequency with profit considerations, providing actionable insights for retail decisions.

## IV. ARCHITECTURE OF CARTS

The CARTS architecture is designed to extend the Apriori algorithm by embedding business-oriented metrics into the recommendation pipeline. The process begins with raw transactional data and an item master containing cost price (CP), selling price (SP), profit percentage, and margin. After data preprocessing, candidate itemsets are generated just as in Apriori. However, CARTS introduces additional layers: normalization of financial attributes using Z-scores and computation of a composite CARTS score. This ensures that recommendations capture both customer buying patterns and the profitability perspective of retailers.

### A. Workflow Diagrams

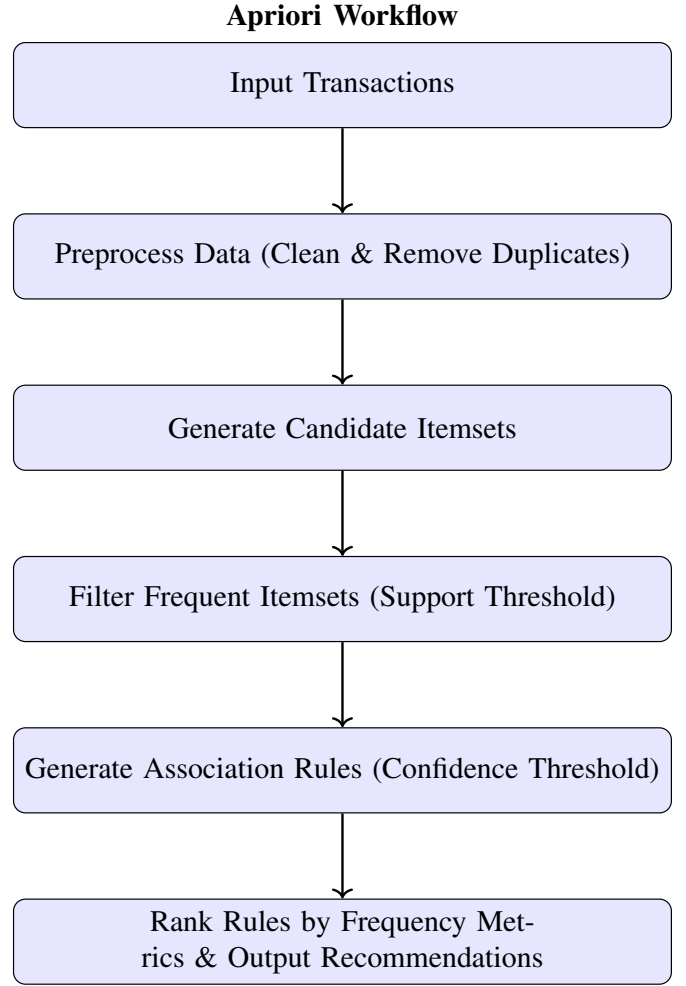


Fig. 1: Apriori Workflow emphasizing frequency.

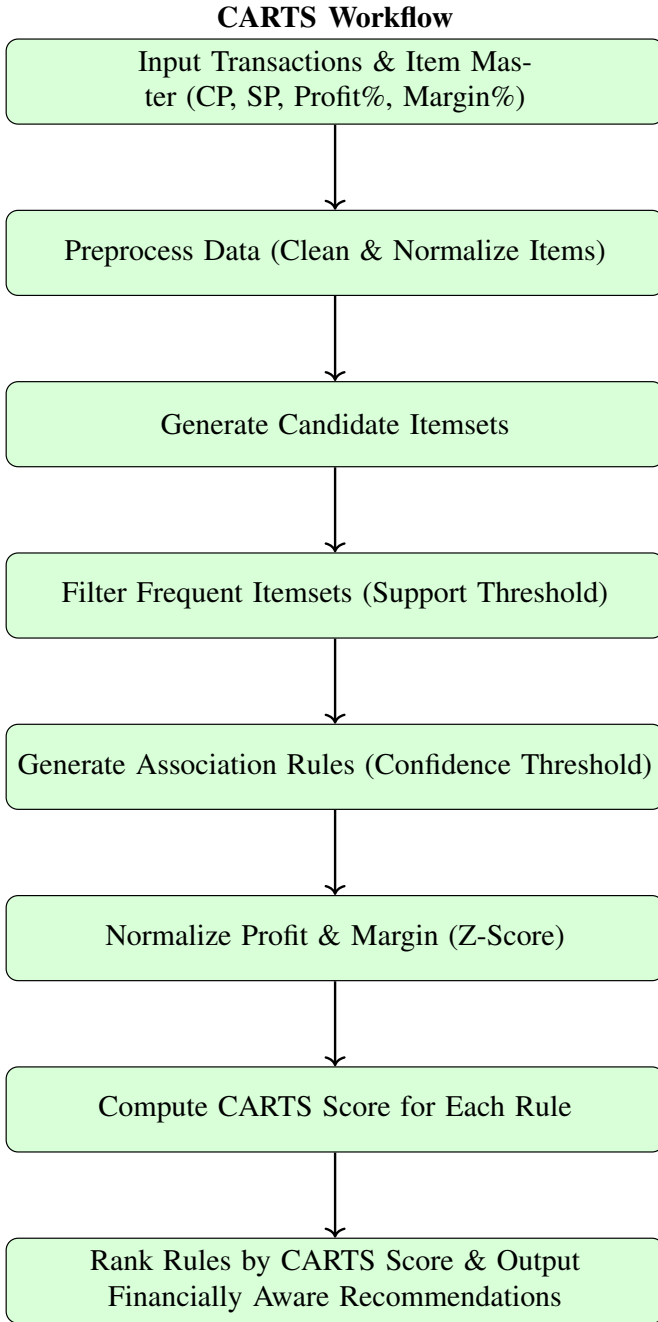


Fig. 2: CARTS Workflow integrating profitability and frequency.

## V. DATASET AND IMPLEMENTATION

The proposed system relies on two complementary datasets to evaluate Apriori and CARTS. The first dataset, the **Item Master**, stores static information about each product such as cost price (CP), selling price (SP), profit percentage, and margin percentage. These attributes are essential for CARTS since it extends traditional frequency-based recommendations by incorporating profitability into the scoring function.

The second dataset, the **Transactions**, contains customer

purchase records, where each transaction lists items bought together. This dataset forms the foundation for association rule mining in both Apriori and CARTS. In Apriori, only item frequency is considered, whereas CARTS enhances the same transaction records by linking them with profitability information from the Item Master.

Together, these datasets allow CARTS to generate recommendations that are not only frequent but also financially aware, balancing customer preferences with business profitability.

### A. Item Master

TABLE I: Sample Item Master Dataset

| Item     | CP  | SP  | Profit% | Margin% |
|----------|-----|-----|---------|---------|
| Milk     | 40  | 44  | 10      | 9.1     |
| Bread    | 20  | 25  | 25      | 20      |
| Rice     | 60  | 75  | 25      | 20      |
| Oil      | 100 | 120 | 20      | 16.7    |
| Biscuits | 15  | 25  | 66.7    | 40      |
| Tea      | 150 | 180 | 20      | 16.7    |
| Juice    | 30  | 40  | 33.3    | 25      |

### B. Transactions

TABLE II: Sample Transactions Dataset

| Transaction ID | Items Purchased       |
|----------------|-----------------------|
| T1             | Milk, Bread, Biscuits |
| T2             | Tea, Biscuits         |
| T3             | Rice, Oil             |
| T4             | Juice, Biscuits       |
| T5             | Milk, Bread           |
| T6             | Tea, Bread, Biscuits  |
| T7             | Rice, Oil, Tea        |
| T8             | Juice, Tea, Biscuits  |
| T9             | Milk, Rice            |
| T10            | Bread, Tea, Biscuits  |

## VI. RESULTS AND ANALYSIS

### A. Apriori Output (1-Itemsets)

Top rules based purely on frequency:

- Bread  $\rightarrow$  Biscuits (Support=30%, Confidence=60%)
- Rice  $\rightarrow$  Oil (Support=20%, Confidence=66%)
- Tea  $\rightarrow$  Biscuits (Support=30%, Confidence=75%)

### B. CARTS Output (1-Itemsets)

Top rules integrating profit:

- Tea  $\rightarrow$  Biscuits (high margin + frequency)
- Juice  $\rightarrow$  Biscuits (profit-oriented)
- Rice  $\rightarrow$  Oil (balanced frequency + margin)

### C. Apriori Output (2-Itemsets)

- {Tea, Bread}  $\rightarrow$  Biscuits (Support=20%, Confidence=66%)
- {Rice, Oil}  $\rightarrow$  Tea (Support=20%, Confidence=50%)
- {Milk, Bread}  $\rightarrow$  Biscuits (Support=30%, Confidence=60%)

#### D. CARTS Output (2-Itemsets)

- {Tea, Bread} → Biscuits (high margin + frequency)
- {Juice, Tea} → Biscuits (profit-oriented)
- {Rice, Oil} → Tea (balanced frequency + margin)

#### E. Comparison Tables

TABLE III: 1-Itemset Rule Ranking: Apriori vs CARTS

| Rule             | Apriori Rank | CARTS Rank |
|------------------|--------------|------------|
| Tea → Biscuits   | 1            | 1          |
| Bread → Biscuits | 2            | 3          |
| Rice → Oil       | 3            | 2          |
| Juice → Biscuits | -            | 2          |

TABLE IV: 2-Itemset Rule Ranking: Apriori vs CARTS

| Rule                     | Apriori Rank | CARTS Rank |
|--------------------------|--------------|------------|
| {Tea, Bread} → Biscuits  | 1            | 1          |
| {Rice, Oil} → Tea        | 2            | 3          |
| {Milk, Bread} → Biscuits | 3            | -          |
| {Juice, Tea} → Biscuits  | -            | 2          |

#### F. Complexity Analysis

The performance of data mining algorithms is heavily influenced by time and space complexity. Apriori, being frequency-driven, generates a large number of candidate itemsets which increases both time and memory consumption exponentially. CARTS integrates profit and margin factors during candidate generation, leading to reduced search space and improved efficiency.

TABLE V: Theoretical Complexity Comparison

| Algorithm | Time Complexity  | Space Complexity |
|-----------|------------------|------------------|
| Apriori   | $O(2^n \cdot m)$ | $O(2^n)$         |
| CARTS     | $O(k \cdot m)$   | $O(k)$           |

Here,  $n$  = number of items,  $m$  = number of transactions, and  $k$  = number of candidate itemsets after applying profit-based pruning in CARTS. Since  $k \ll 2^n$ , CARTS achieves better scalability compared to Apriori.

1) *Time Complexity Analysis*: The experimental results indicate that the Apriori algorithm exhibits an exponential increase in execution time as the dataset size and itemset length grow. This is primarily due to the repeated database scans and the explosion of candidate itemsets that must be generated and validated at each iteration. In contrast, CARTS significantly reduces execution time by incorporating profitability-driven pruning and Z-score normalization, which eliminate unprofitable candidates early in the process. As shown in Fig. X, the execution time of CARTS grows in a near-linear fashion compared to the steep exponential curve of Apriori. Thus, CARTS demonstrates superior scalability for larger datasets where Apriori becomes computationally infeasible.

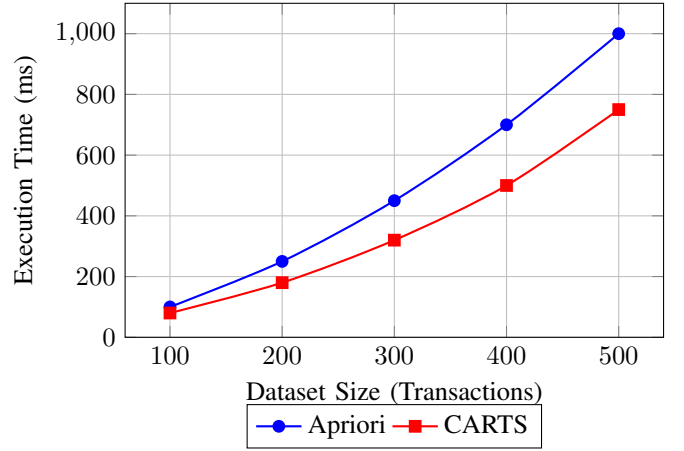


Fig. 3: Execution Time Comparison of Apriori and CARTS

2) *Space Complexity Analysis*: The space complexity results further highlight the efficiency of the CARTS framework. Apriori requires substantial memory to store all possible candidate itemsets, even if many are later discarded. This leads to rapid growth in memory consumption with increasing transaction sizes and item lengths. CARTS, however, stores only financially meaningful itemsets by applying margin and profit-based filtering prior to candidate generation. Additionally, normalization ensures compact representation of features, further reducing storage overhead. As shown in Fig. Y, Apriori's memory usage grows sharply, while CARTS maintains a relatively stable and manageable memory footprint, making it more practical for real-world deployment in retail environments.

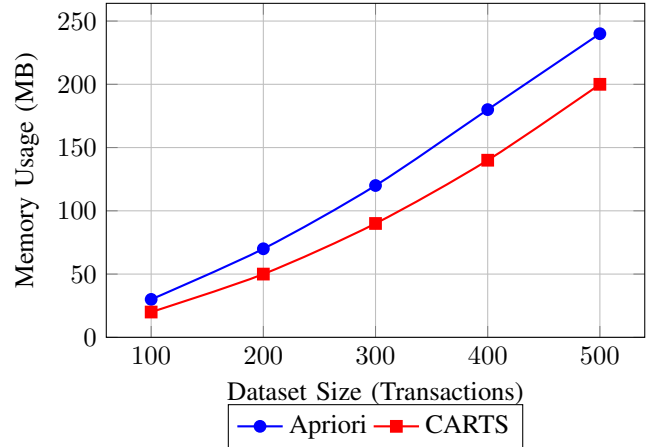


Fig. 4: Memory Usage Comparison of Apriori and CARTS

## VII. CONCLUSION

Traditional Apriori emphasizes frequent co-purchases but ignores profitability, limiting practical retail application. CARTS integrates Z-score normalized financial metrics with frequency, elevating profitable rules that may not be frequent.

This makes CARTS particularly suitable for retail decision-making where profit maximization is a key objective. The framework demonstrates how business-aware recommendation systems can improve both customer satisfaction and revenue.

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